Design of a Firefighting Robot: An Experimental Comparison of Direct Visual and Gyroscope-Assisted Dual-Loop PID

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INTRODUCTION

Problem & Motivation

- Ground obstacles (e.g., thresholds, cables) cause wheel impacts and chassis vibration, displacing the line-of-sight and deflecting the water jet.
- Research Gap: Controller performance under graded obstacle impacts has rarely been systematically evaluated for continuous mobile aiming.

Research Scope

- Our study aims to provide rigorous experimental evidence and controller selection guidelines for engineers designing mobile firefighting robots.
- This Study evaluates Direct Visual vs. Gyroscope-Assisted PID on a custom wheeled firefighting robot.



Fig. 1. Real-world challenges in firefighting robots.

PROTOTYPE SYSTEM

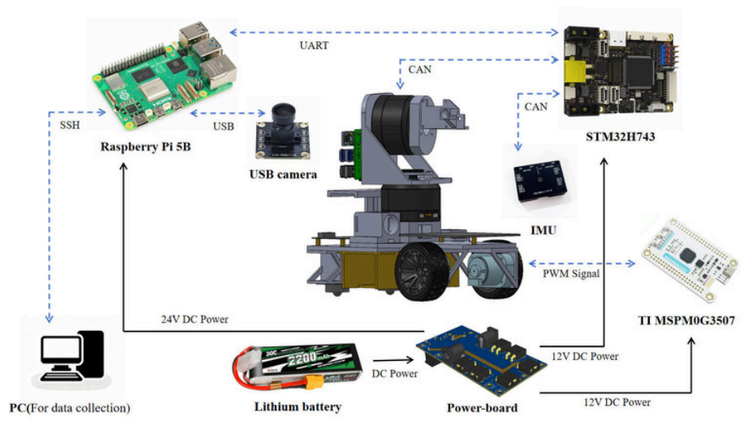


Fig. 2. System architecture of the developed robot.

- **Custom-Built Platform:** A dedicated wheeled robot integrating onboard vision tracking, an inertial-stabilized two-axis gimbal, and embedded chassis control for repeatable benchmark testing.

THEORETICAL ANALYSIS

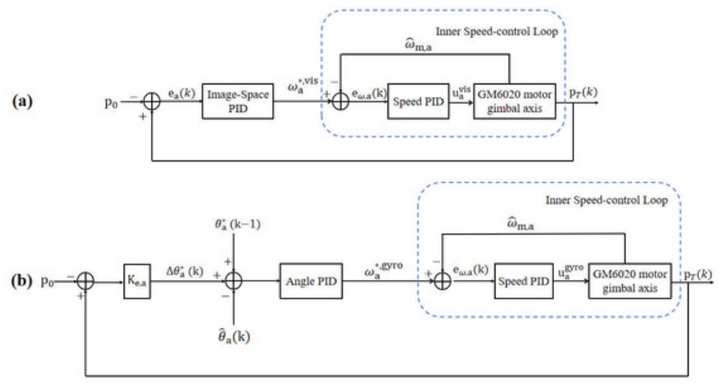


Fig. 3. Control structures of the two PID controllers.

- **Direct Visual Dual-Loop PID:** Directly feeds image pixel errors to command motor speed.
- **Gyroscope-Assisted Dual-Loop PID:** Converts visual errors into angular increments, using a high-rate gyroscope outer loop to reject disturbances between visual updates.

EXPERIMENTAL METHOD

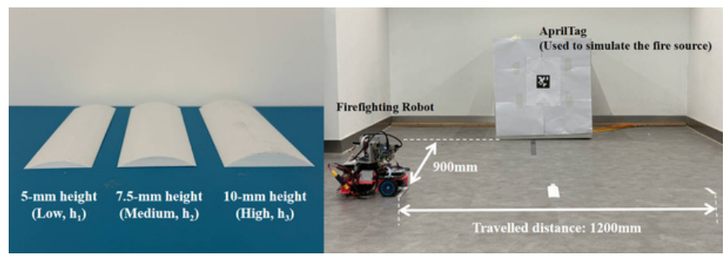


Fig. 4. Obstacle set and experimental layout.

- A repeatable benchmark evaluated the aiming performance of both controllers across flat ground and graded obstacles.

Condition	Fixed value or procedure
Start pose and path	Common marked traversal; 1200 mm
Path–tag perpendicular distance	900 mm
Wheel and nominal linear speed	0.7 rev/s; approx. 0.143 m/s
Obstacle height	0, 5, 7.5, and 10 mm
Wheel radius	32.5 mm
Target geometry	tag36h11; 100 × 100 mm
Effective image-sample rate	Approximately 41 Hz
Power source	Common 6S battery system
Controller parameters	Tuned first, then frozen

Table. 1. Controlled experimental conditions

RESULTS AND DISCUSSION

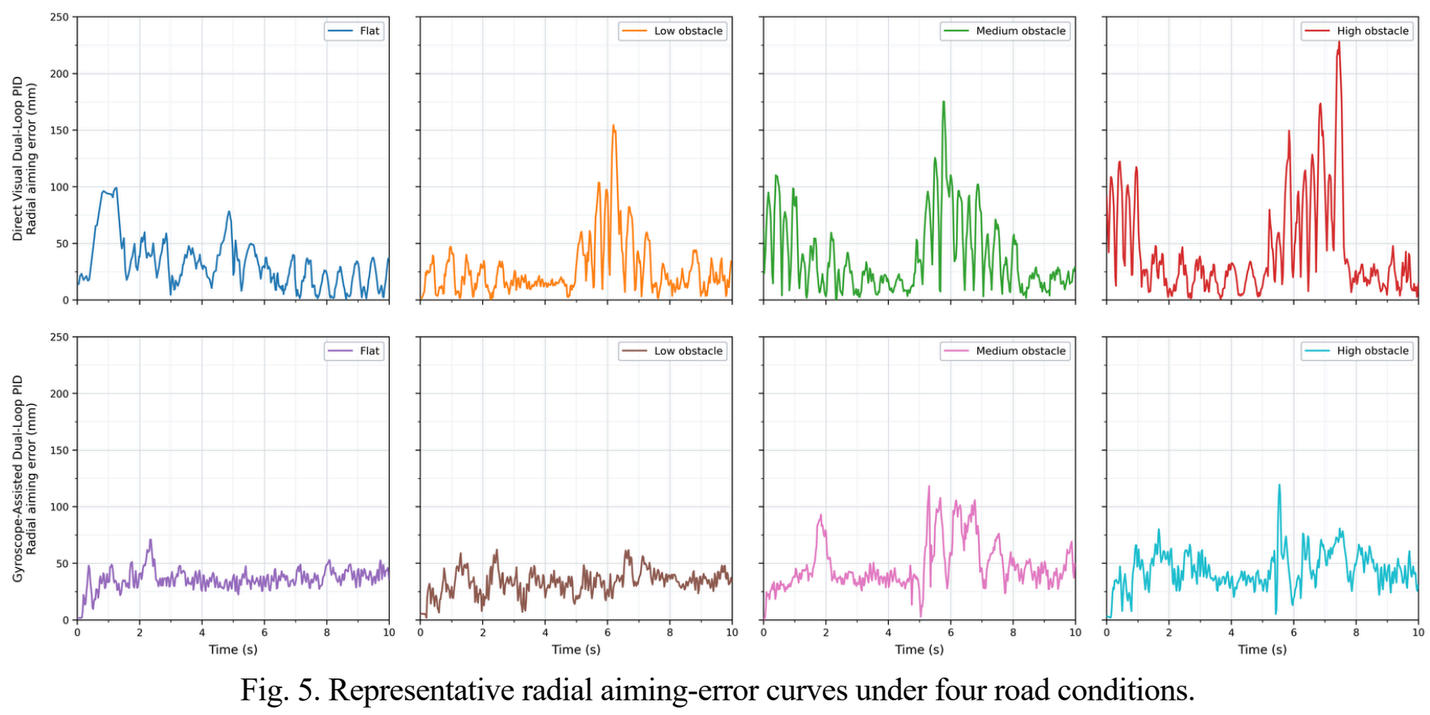


Fig. 5. Representative radial aiming-error curves under four road conditions.

Road	RMSE (mm)	e_{95} (mm)	Max. (mm)	Success (%)
Direct Visual Dual-Loop PID				
Flat	31.34 ± 4.85	62.53 ± 8.60	104.25 ± 31.10	98.04 ± 1.81
Low	31.48 ± 6.90	62.15 ± 16.78	106.80 ± 48.32	97.09 ± 2.60
Medium	47.47 ± 9.12	98.18 ± 11.23	166.91 ± 40.41	86.22 ± 5.38
High	54.42 ± 7.02	108.72 ± 17.97	189.66 ± 38.23	83.73 ± 7.04
Gyroscope-Assisted Dual-Loop PID				
Flat	36.06 ± 2.05	50.71 ± 4.82	74.26 ± 2.99	99.92 ± 0.14
Low	38.37 ± 4.01	61.05 ± 11.49	78.13 ± 20.50	98.46 ± 2.67
Medium	53.73 ± 1.39	90.90 ± 4.13	126.52 ± 7.54	86.51 ± 0.56
High	46.98 ± 1.88	76.45 ± 5.15	120.46 ± 5.02	94.35 ± 2.92

Table. 2. Measured group results (three trials per group)

- **Experimental Insight:** Experimental results show that gyroscope-assisted control suppresses transient peak aiming errors during obstacle impacts, while direct visual control yields lower overall RMSE. This confirms gyroscope assistance primarily mitigates peak disturbances rather than minimizing cumulative error.

CONCLUSION

- **Key Findings:** Gyroscope-Assisted Dual-Loop PID consistently suppresses peak pointing errors across all road conditions, serving primarily to reject transient disturbance spikes rather than reduce overall cumulative quadratic error (RMSE)
- **Design Implication:** When limiting peak pointing error is the primary design objective, Gyroscope-Assisted Dual-Loop PID is preferable to Direct Visual Dual-Loop PID.
- **Future Work:** Future work will evaluate the system using real water discharge and actual fire sources on more complex terrain.

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